

Flexible Emission Models for Hidden Markov Models in Financial Regime Detection - Project Proposal

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Financial markets exhibit complex dynamics, which are characterized by periods of high and low volatility, bullish and bearish trends, and changing market conditions. These periods are commonly referred to as market regimes [1], and they are widely used to describe different phases of market behavior. Although broad bull and bear markets are often visually apparent, financial markets frequently exhibit more subtle latent regimes that cannot be directly observed. Identifying these hidden regimes can improve risk management, portfolio allocation, and market forecasting. This motivates the idea of inferring these latent regimes using unsupervised learning methods.

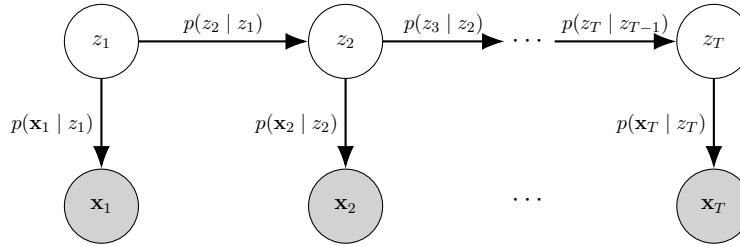


Figure 1: Hidden Markov model as a directed graph.

Hidden Markov Models provide a probabilistic framework for modeling latent market regimes [2]. In an HMM used in our context, it is assumed that market observations $\mathbf{x}_{1:T}$ are generated from hidden market regimes $z_{1:T}$ that evolve according to a first-order Markov chain, it can be represented as a directed graph shown in Figure 1. The transition distribution $p(z_t|z_{t-1})$ captures the temporal evolution of latent regimes, while the emission distribution $p(\mathbf{x}_t|z_t)$ specifies how each hidden regime gives rise to the observed market features. For the purpose of regime detection, we're interested in inferring the corresponding sequence of hidden regimes given a sequence of market features, or more generally the posterior distribution $p(z_{1:T}|\mathbf{x}_{1:T}) = p(\mathbf{x}_{1:T}|z_{1:T})p(z_{1:T})/p(\mathbf{x}_{1:T})$. Therefore, the quality of the regime detection has high dependence on whether the chosen emission model sufficiently represents the distribution found in the data.

Despite their widespread use, traditional HMMs typically assume Gaussian emission distributions (i.e. $\mathbf{x}_t|z_t = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$). This assumption is often unrealistic for financial data, as they commonly exhibit heavy tails, skewness, volatility clustering, and other non-Gaussian characteristics [3]. Consequently, Gaussian emission models are highly likely to inadequately represent market observations and reduce the quality and validity of the inferred regimes.

We propose to extend flexible emission models to HMM to improve market regime detection. Specifically, we consider a progression of emission models with increasing expressive power: Gaussian, Gaussian Mixture, and Normalizing Flow-based emissions. Gaussian Mixture emissions increase modeling flexibility by representing multi-modal observation distributions, while Normalizing Flows are a modern deep learning approach capable of modeling complex skewed and heavy-tailed financial observations by transforming a simple distribution into the shape of our target distribution using a series of invertible and differentiable functions [4]. This project seeks to determine whether increasingly expressive emission distributions improve the predictive performance and interpretability of HMMs for financial market regime detection.

Our candidate data set is the S&P 500 daily index from 2000-2026, which is publicly available from Yahoo Finance. The proposed models will be evaluated using out-of-sample log-likelihood, AIC/BIC, regime persistence, posterior confidence, and economic interpretability. We will characterize each inferred regime by its statistical properties (e.g., mean return and volatility) and compare its alignment with major historical market events such as the 2008 Global Financial Crisis and the COVID-19 market crash. As an optional downstream evaluation, we will assess a simple regime-based trading strategy using the Sharpe ratio and maximum drawdown for risk evaluation across all 3 models.

References

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